

Engineering Decision Report

NovaloT FleetGuard — Cold-Chain Container Monitoring Device

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This report explains the engineering reasoning behind the hardware and architecture selections made for the FleetGuard container monitoring device, the trade-offs considered, what would change for a production deployment, known risks and limitations, and planned future improvements. It should be read alongside **docs/Hardware_Design.md**, **docs/System_Architecture.md**, and **docs/Firmware_Architecture.md**, which contain the full technical detail this report summarizes and justifies.

1. Why This Hardware Was Selected

1.1 Processing Unit — ESP32-S3-WROOM-1

The ESP32-S3 was chosen over alternatives such as the STM32 family, RP2040, or a plain Arduino board because it uniquely satisfies three requirements at once: sufficient compute headroom for local classification logic running under FreeRTOS, integrated Wi-Fi/BLE for the future connectivity requirement without adding a second radio module, and a mature, well-documented ecosystem that reduces development risk for a first-of-its-kind MVP. STM32 parts offer excellent low-power and real-time characteristics but would require a separate wireless module, adding board area, cost, and integration risk that isn't justified when the workload (periodic sampling, threshold checks, local logging) is not real-time-critical in the hard sense. The RP2040 lacks integrated wireless entirely. The dual-core architecture also allows sensor/alert servicing to be separated from future networking tasks on FreeRTOS without them contending for the same core.

1.2 Environmental Sensor — Sensirion SHT41-AD1B

Temperature and humidity are the two mandatory monitored parameters. The SHT41 was selected for its factory calibration (removing the need for field calibration across thousands of deployed units), its integrated heater (which drives off condensation — a real risk in a refrigerated, humid environment), and its accuracy specification, which is appropriate for perishable-goods compliance reporting. A cheaper DHT22-class sensor was considered and rejected: its accuracy and long-term drift characteristics are not suitable for a device whose readings may need to stand up to a customer dispute about spoiled cargo.

1.3 Optional Sensors

Door status (reed switch), battery voltage (resistive divider + ADC), ambient light (BH1750), and vibration (LIS3DH) were added because each provides low-cost context that materially improves the operator's ability to diagnose *why* a temperature excursion happened — a door left open explains a temperature spike in a way a bare temperature graph cannot. All four use well-supported, inexpensive, digital interfaces (GPIO/I2C/ADC) that don't meaningfully add to BOM cost or firmware complexity at fleet scale.

1.4 Power Subsystem

The device is assumed to run from 12V/24V vehicle power. Reverse polarity protection (P-channel MOSFET), a resettable PTC fuse, and a TVS diode were included specifically because vehicle electrical systems are electrically noisy and prone to install-time wiring mistakes at scale — for a fleet of thousands of units installed by many different technicians, protecting against a reversed connector is cheaper than the warranty and support cost of not doing so.

1.5 Local Storage

A hybrid of ESP32 NVS (configuration), external SPI flash (event buffer), and an industrial-grade MicroSD card (long-term history) was chosen over a single storage medium to balance write endurance against capacity: the SPI flash buffer absorbs frequent small writes without wearing out the SD card's limited write cycles, while the SD card provides the bulk capacity needed for weeks of unattended operation between depot visits.

2. Why This Architecture Was Selected

2.1 Layered System Architecture

The system is split into six layers (Sensing, Data Acquisition, Decision Engine, Event Logging, Alerting, Future Connectivity) so that each layer has a single, testable responsibility and can evolve independently. In particular, isolating sensor access behind a Data Acquisition layer means the Decision Engine, Event Logger, and Alert Manager never call hardware APIs directly — this is what allowed the entire pipeline to be demonstrated as a software simulation (Task 4, Option B) without any physical board, and it is what will let the same logic run unmodified against real I2C/GPIO/ADC drivers in production firmware.

2.2 Worst-Case-Wins Classification

Rather than a single combined "health score," each monitored parameter is classified independently against a two-band threshold (Normal / Warning / Critical), and the overall device condition is the worst individual classification. This was a deliberate, conservative choice: an average or weighted score could let one badly out-of-range parameter be masked by others being fine, which is unacceptable for a safety/compliance-relevant cold-chain product. The two-band structure (rather than a single cutoff) exists to avoid alert chattering right at a boundary and to give operators lead time between Warning and Critical.

2.3 Device State Machine Separate From Condition

Device operating state (BOOT / RUNNING / DEGRADED / FAULT / CONFIG) is modeled separately from environmental condition (Normal / Warning / Critical / Fault). This distinction matters operationally: "the sensor stopped responding" and "the cargo is too warm" require different responses from a technician versus an operations manager, and conflating them in one status field would make the device harder to diagnose in the field.

2.4 Crash-Safe Event Logging

Every event write is flushed and fsync'd before being considered committed, so an abrupt power loss (a realistic event on vehicle power) can lose at most the one in-flight write, never previously committed history. This directly addresses the Reliability non-functional requirement and was validated in the prototype by a scripted power-loss/restart test (see firmware/main.py, Phase 9).

3. Trade-offs Considered

- **Option A (physical hardware) vs Option B (simulation):** Option B was chosen because no physical ESP32-S3 board, sensors, or lab equipment were available in this environment within the assignment's time window. The trade-off is that timing, real bus behaviour, and physical failure modes are not hardware-validated — mitigated by keeping the simulation's module boundaries and logic identical to the intended production firmware, so the design itself is still verifiable.
- **Single MCU edge device vs MCU + separate radio/gateway:** A single ESP32-S3 per container was chosen over a per-container sensor node reporting to a shared gateway, because a self-contained device with local alerting works even when a container is isolated from other units (e.g., in transit, doors closed, no line-of-sight to a gateway) — important since local alerting must work with zero connectivity.
- **No local display vs adding a display:** A display was rejected in favour of LED + buzzer + serial diagnostics, trading richer on-device information for lower cost, lower power draw, and less exposure to a harsh environment (moisture, vibration, glass/screen damage risk) across a fleet of thousands.

- **Hybrid flash+SD storage vs SD-only:** SD-only would be simpler but exposes frequent small event writes directly to the SD card's limited write-cycle endurance; the SPI flash buffer trades a small amount of extra BOM cost and firmware complexity for meaningfully better long-term reliability at fleet scale.
- **Threshold-based classification vs a statistical/ML approach:** A deterministic, configurable threshold model was chosen over an ML-based anomaly detector because it is auditable, explainable to a non-technical operator, and doesn't require training data that doesn't exist yet for a first-of-its-kind deployment. This trade-off is revisited under Future Improvements (predictive maintenance).

4. What Would Change For Production Deployment

- **Firmware:** port the simulation's module logic to ESP-IDF/FreeRTOS C, replacing simulated sensor classes with real I2C/GPIO/ADC drivers behind the same interface boundaries.
- **Hardware validation:** full schematic capture and PCB layout (this submission provides component selection and a block diagram, not a schematic), plus EMI/EMC and automotive electrical compliance testing given the 12V/24V vehicle power environment.
- **Security:** secure boot and firmware signing to protect firmware integrity, per-device identity/credentials (e.g., provisioned at manufacturing time) for fleet management and to prevent unauthorized configuration changes, and TLS for any future cloud communication.
- **OTA updates:** a staged OTA rollout mechanism (canary devices first, automatic rollback on boot failure) is essential once thousands of units are deployed in the field and cannot be physically re-flashed.
- **Manufacturing test:** an automated end-of-line test procedure (sensor presence/communication check, power-on self-test, provisioning) to catch defective units before shipment.
- **Enclosure & certification:** an IP-rated enclosure suitable for a refrigerated, humid, vibration-prone environment, plus applicable regulatory certification (e.g., FCC/CE for the radio, automotive EMC standards).
- **Cloud integration:** implement the MQTT/cloud ingest path the architecture already reserves space for, including device provisioning and fleet-scale device management.

5. Risks and Limitations

- **No hardware-in-the-loop validation:** real sensor failure modes (e.g., I2C bus lockup, intermittent connector faults) may behave differently than the simulated fault injection used here.
- **Single point of environmental sensing:** one SHT41 per container means a physical sensor failure removes the primary monitored signal for that unit until repaired; the design falls back to last-known-good data but cannot indefinitely substitute for a working sensor.
- **Threshold strategy is generic:** default thresholds (2-8°C / 30-70% RH) target chilled goods; different cargo types (frozen, pharmaceutical) need their own validated threshold sets before deployment, which the configuration system supports but does not pre-populate.
- **Security is designed-for, not implemented:** the current prototype has no authentication on configuration changes; this is acceptable for a design exercise but is a real gap before any field deployment.

- **Vehicle power quality is assumed, not measured:** the power subsystem is designed against typical automotive transient specifications from datasheets, not against measured data from an actual target vehicle fleet.

6. Future Improvements

- Secure OTA firmware update architecture with staged rollout and rollback (Bonus 1).
- Predictive maintenance: trend battery voltage, sensor read latency, and vibration signatures over time to flag a likely refrigeration unit failure before a critical threshold is crossed, rather than only reacting after the fact (Bonus 6).
- A lightweight web/mobile dashboard surfacing live device status and historical events per container, aimed at non-technical operations managers (Bonus 5).
- A remote fleet-management architecture (device registry, bulk configuration push, health dashboard) designed explicitly for 50,000+ device scale (Bonus 4).
- A documented manufacturing test procedure for end-of-line validation before units ship (Bonus 3).
- Formal schematic capture and PCB layout once component selection is validated against real procurement/lead-time constraints.

End of report.